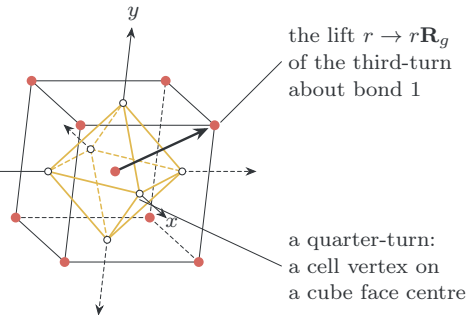


stripped to
its core



$\mathcal{F}$ cut open: the core is $SO(3)$,
 $\mathbf{X}_0$ in its body frame

the wall the nine vibrations
... twelve sheets, one per rotation of T

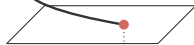
$r\mathbf{R}_g^2$



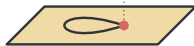
$r\mathbf{R}_g$



r



$SO(3)/T$



ℓ_g

the covering $SO(3) \rightarrow SO(3)/T$ as in Figure 11;

the lift of ℓ_g runs from sheet r to sheet $r\mathbf{R}_g$

$SO(3)$ in Rodrigues coordinates, direction the axis
and distance $\tan \frac{\theta}{2}$: the centre, the cube of third-turns,
the cell $\mathcal{U}$ at $q = 0$, the half-turns at infinity

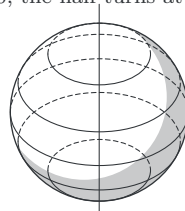
$K = 2$

$K = 1$

$K = 0$

$K = -1$

$K = -2$



${}^2\mathbf{E}$

${}^1\mathbf{E}$

$\mathbf{A}$

${}^2\mathbf{E}$

${}^1\mathbf{E}$

Figure 12's rings of K , $J = 2$, z along bond 1:

the third-turn multiplies ring K by ω^K ,

so $\mathbf{A}$ when $3 \mid K$, else ${}^1\mathbf{E}$ or ${}^2\mathbf{E}$